

3.4 BANDWIDTH CONTROLS OPTIMIZATION ACCESS TO HIGH PRECISION

The construction’s nearly width-independent relative bandwidth $\lambda = \gamma h$ requires $\gamma = \Theta(W)$, yet trained mean slopes fall short (Figure 4). Even though larger widths admit smaller λ at comparable construction error (Figure 1b), small slopes can delay readout learning by weakening target-relevant kernel directions. We quantify this delay by replacing Section 3.1’s direct least-squares solve with GD on fixed features. This connects prior analysis of spectral bias (Xu et al., 2020; Tancik et al., 2020) to readout optimization in the quasi-interpolant dictionary.

Theorem 3.2 (Slope-dependent output-error lower bound). *For the fixed-center construction of Section 3.1 with common slope $\gamma > 0$, let $\mu_1 \geq \mu_2 \geq \dots \geq 0$ be the normalized readout-kernel eigenvalues. For a nonzero sampled target, readout GD from zero with $\eta = 1/(2\mu_1)$ satisfies*

$$\frac{\|f - q_n\|_{2,Z}^2}{\|f\|_{2,Z}^2} \geq \sum_i p_i(\gamma) \left(1 - \frac{\bar{\rho}_i(\gamma)}{2}\right)^{2n}, \quad n \geq 0. \quad (1)$$

Here $\|\cdot\|_{2,Z}$ is the training-sample RMS norm, p_i is the fraction of sampled target energy projected onto kernel eigenvector i , and $\mu_i/\mu_1 \leq \bar{\rho}_i(\gamma) \leq 1$. The gamma-dependent upper bounds $\bar{\rho}_i$ come from the spectral Theorem A.1 in Appendix A.

Proof sketch. The readout kernel contains normalized inner products of feature vectors at pairs of training points. To bound its spectrum, we separate the constant sample direction and approximate the sum over centers by an integral. This approximation depends explicitly on $d \coth(\gamma d)$, where d separates those points. Bounding the finite-center corrections gives $\bar{\rho}_i(\gamma)$; standard GD eigenmode decay then yields the output-error bound.

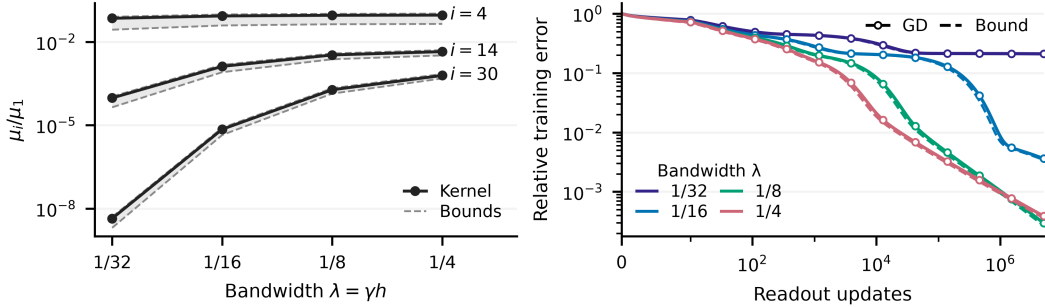


Figure 3: **Small slopes weaken kernel directions and delay readout learning.** Width $W = 512$; target $f(x) = [\sin(2\pi x) + \frac{1}{2}\sin(6\pi x) + \frac{1}{4}\sin(14\pi x)]/\sqrt{21/32}$ on $[-1, 1]$. **Left:** Kernel ratios μ_i/μ_1 (dots) and intervals from Theorem A.1 (dashed boundaries, shading) for three ranks carrying substantial target energy. Lines connect ordered ranks; eigenvectors change with bandwidth. **Right:** Executed frozen-readout GD errors (solid, hollow markers) and square roots of Theorem 3.2’s lower bounds (dashed). The error bound combines theorem-derived rate bounds with all resolved target projections onto the frozen kernel’s eigenvectors. Errors are relative training RMS through five million updates; geometry and protocol are in Appendix B.

The spectral bounds of Theorem A.1 track how bandwidth changes target-relevant ratios μ_i/μ_1 (Figure 3, left). GD’s step size is limited by the largest eigenvalue μ_1 . At small bandwidths, target-relevant eigenvalues are much smaller, so the corresponding error components decay slowly. Theorem 3.2 converts these rate bounds into output-error lower bounds that closely track executed GD (Figure 3, right): small slopes delay accuracy already available in the dictionary.

Joint training can change this spectrum by moving the slopes and centers. We next examine whether it acquires useful features within the training budget.

3.5 WHY JOINT TRAINING ACQUIRES USEFUL FEATURE SCALES SLOWLY

Allowing the features to evolve leaves a substantial accuracy gap over the tested training budget. After five million updates on mixed sine at $W = 512$, joint Adam reaches median relative training error 1.81×10^{-3} , compared with 6.62×10^{-7} using the supplied $\lambda = 1/4$ dictionary; joint GD remains at 0.244 (Figure 4b). Numerically resolved readout refits barely improve the learned Adam dictionary, and increasing its slopes at fixed centers does not close the gap (Appendix D). These interventions indicate that useful slopes and centers must be acquired together.

We quantify one mechanism that limits this acquisition. In the small-parameter regime, $\tanh$ ’s leading non-affine term is cubic, so its target-driven growth depends on the target’s low-degree polynomial projection. Bounding this coupling, higher-order contributions, and population concentration controls the growth of slopes, biases, and readouts. The resulting bound on non-affine output yields an output-error lower bound.

Theorem 3.3 (Output error under slow feature acquisition). *Let $f \neq 0$ be the sampled target and $f_\perp$ its component orthogonal to affine functions. Under the population-concentration, target-coupling, and coarse-tracking bounds of Appendix C, joint GD satisfies*

$$\frac{\|f - q_n\|_{2,Z}}{\|f\|_{2,Z}} \geq \frac{[\|f_\perp\|_{2,Z} - \mathcal{Q}_n(B_n)]_+}{\|f\|_{2,Z}}. \quad (2)$$

Here B_n is an explicit scalar upper bound on the joint norm of hidden slopes, biases, and readouts, and $\mathcal{Q}_n(B_n)$ bounds their non-affine output. The same comparison gives $\bar{\lambda}_n \leq hB_n/\sqrt{W}$, with reference spacing h and $\bar{\lambda}_n = W^{-1} \sum_j h|\gamma_{j,n}|$. The recurrence, coefficients, and precise conditions are given in the appendix.

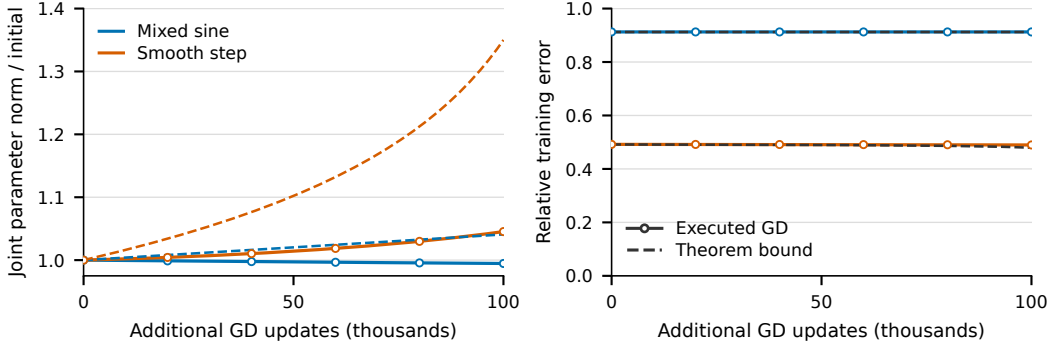


Figure 5: **Limited population growth leaves a substantial output-error floor. Left:** Joint parameter norm and its upper bound B_n , normalized by the initial norm. **Right:** Executed GD relative training RMS error and the theorem’s lower bound. Solid curves and markers show GD; dashed curves show bounds. Archived $W = 705$, seed-30 continuations start after 20,000 updates and run 100,000 more at step size 0.002. Mixed sine is the target used in Figure 4; smooth step has the largest growth allowance among the six principal targets. Bounds use structural coefficients evaluated along the saved trajectories.

The conditional bounds closely track the separate GD continuation experiments: across ten continuations spanning six target families, the output-error lower bound captures at least 97.7% of executed error at every stored diagnostic time (Figure 5; Appendix D). The structural coefficients are evaluated along these trajectories. The longer Adam experiments establish a corresponding empirical acquisition gap; the theorem concerns GD.

QUILL supplies the feature geometry and readout directly, avoiding this acquisition stage. We next compose these constructed primitives into arithmetic circuits.

Review copy. Figure 4’s three-panel revision is pending and is not included here. The Section 3.5 theorem and proof use completed evidence.

A Explicit spectral bounds and proof

The main theorem concerns the actual finite tanh dictionary. An auxiliary center integral exposes its gamma dependence; lattice and exterior-center allowances transfer its spectrum to that dictionary. This appendix specifies each approximation and the resulting endpoints. The scale-dependent spectral bias is related to [Xu et al. \(2020\)](#); [Tancik et al. \(2020\)](#); the argument below quantifies it for the finite quasi-interpolant dictionary.

Finite setup. Let $z_1, \dots, z_M$ be distinct training points with $M \geq 2$, W consecutive centers on $c_0 + h\mathbb{Z}$ cover their interval, and $\gamma > 0$. Define

$$\Phi_\gamma = [\mathbf{1}, (\tanh(\gamma(z_a - x_j)))_{a,j}], \quad K_\gamma = \frac{1}{M} \Phi_\gamma \Phi_\gamma^\top, \quad y = (f(z_a))_{a=1}^M \neq 0. \quad (3)$$

Readout GD minimizes $\|\Phi_\gamma w - y\|^2 / (2M)$. Write its descending kernel eigenpairs as (μ_i, u_i) and define

$$\rho_i = \mu_i / \mu_1, \quad p_i = |u_i^\top y|^2 / \|y\|^2, \quad e_n^2 = \frac{\|\Phi_\gamma w_n - y\|^2}{\|y\|^2} = \frac{\|f - q_n\|_{2,Z}^2}{\|f\|_{2,Z}^2}. \quad (4)$$

The energy weights sum to one, including any exact nullspace component. The theorem permits arbitrary distinct samples; the reported experiments use uniform samples. Write $v_0 = \mathbf{1} / \sqrt{M}$ and let Q have orthonormal columns spanning $v_0^\perp$. This projection removes the common constant sample pattern, not any target assumption. Put $t_\gamma(c) = (\tanh(\gamma(z_a - c)))_a$.

The gamma-dependent correlation model. Its center-integrated, mean-zero kernel is

$$H_\gamma = \frac{1}{hM} \int_{\mathbb{R}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q dc = -\frac{2}{hM} Q^\top [d_{ab} \coth(\gamma d_{ab})]_{a,b} Q, \quad d_{ab} = z_a - z_b. \quad (5)$$

The diagonal of the bracketed matrix is $1/\gamma$. Choose a finite set $\mathcal{F}$ of exterior lattice centers absent from the actual dictionary, leaving two unretained tails starting at $c_L < \min z_a$ and $c_R > \max z_a$. Subtract their retained contribution:

$$T_\gamma = \frac{1}{M} \sum_{c \in \mathcal{F}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q, \quad S_\gamma = H_\gamma - T_\gamma. \quad (6)$$

Let $\beta_1 \geq \dots \geq \beta_{M-1}$ be the eigenvalues of S_γ . For any $0 < \theta < \pi/2$, the lattice and remaining-tail allowances are

$$\delta_\gamma = \frac{8\gamma \|z - \bar{z}\mathbf{1}\|^2 \sec^4 \theta}{3hM [\exp(2\pi\theta/(\gamma h)) - 1]}, \quad \bar{z} = M^{-1} \sum_a z_a, \quad (7)$$

$$\tau_\gamma = \frac{4[e^{-4\gamma(c_R - z_{\max})} + e^{-4\gamma(z_{\min} - c_L)}]}{1 - e^{-4\gamma h}}. \quad (8)$$

They give the matrix enclosure

$$S_\gamma - (\delta_\gamma + \tau_\gamma)I \preceq Q^\top K_\gamma Q \preceq S_\gamma + \delta_\gamma I. \quad (9)$$

Relative-eigenvalue endpoints. Normalize using scalar bounds $\ell_\gamma \leq \mu_1 \leq L_\gamma$. The following require only matrix-vector products with the finite dictionary:

$$\ell_\gamma = v_0^\top K_\gamma v_0 \geq 1, \quad b_\gamma = \|Q^\top K_\gamma v_0\|, \quad c_\gamma = \beta_1 + \delta_\gamma, \quad (10)$$

$$L_\gamma = \frac{\ell_\gamma + c_\gamma + \sqrt{(\ell_\gamma - c_\gamma)^2 + 4b_\gamma^2}}{2}. \quad (11)$$

Set $\underline{\rho}_1 = \bar{\rho}_1 = 1$ and $\beta_M = 0$.

Theorem A.1 (Slope-dependent relative-eigenvalue bounds).

$$\boxed{\rho_i = \frac{[\beta_i - \delta_\gamma - \tau_\gamma]_+}{L_\gamma}, \quad \bar{\rho}_i = \min\left\{1, \frac{\beta_{i-1} + \delta_\gamma}{\ell_\gamma}\right\}, \quad 2 \leq i \leq M.} \quad (12)$$

For the finite setup above, these endpoints enclose every kernel ratio: $\underline{\rho}_i \leq \mu_i/\mu_1 \leq \bar{\rho}_i$. Exact zeros known from $\text{rank } K_\gamma \leq W + 1$ may have both endpoints set to zero.

No target or optimization trajectory enters these spectral bounds. The upper endpoints supply Theorem 3.2's output-error lower bound.

A.1 DERIVATION OF THE EXPLICIT GAMMA DEPENDENCE

The identity $1 - \tanh u \tanh v = \coth(u - v)(\tanh u - \tanh v)$ gives

$$\int_{\mathbb{R}} [1 - \tanh(\gamma(z_a - c)) \tanh(\gamma(z_b - c))] dc = 2d_{ab} \coth(\gamma d_{ab}). \quad (13)$$

Multiplication by $Q^\top$ and Q removes the constant term and proves (5). The integral is finite after projection because $Q^\top t_\gamma(c)$ decays at both ends.

The smoothing interpretation can also be made exact. With $s_\gamma(x) = \frac{\gamma}{2} \text{sech}^2(\gamma x)$, $\tanh(\gamma \cdot) = s_\gamma * \text{sign}$. Its Fourier multiplier is

$$A_\gamma(\omega) = \widehat{s}_\gamma(\omega) = \frac{\pi|\omega|/(2\gamma)}{\sinh(\pi|\omega|/(2\gamma))}, \quad A_\gamma(0) = 1. \quad (14)$$

For completeness, substitution $r = e^{2\gamma x}$ in the Fourier transform gives $\int_0^\infty r^{-i\omega/(2\gamma)}(1+r)^{-2} dr$. The beta integral evaluates this as $\Gamma(1 - i\omega/(2\gamma))\Gamma(1 + i\omega/(2\gamma))$, which equals (14) by the reflection formula. For $u = Qv$, the step combination $\sum_a u_a \text{sign}(c - z_a)$ has compact support, and its derivative is $2 \sum_a u_a \delta_{z_a}$. Parseval gives

$$v^\top H_\gamma v = \frac{2}{\pi h M} \int_{\mathbb{R}} \frac{A_\gamma(\omega)^2}{\omega^2} \left| \sum_a (Qv)_a e^{-i\omega z_a} \right|^2 d\omega. \quad (15)$$

The apparent zero-frequency singularity is removable since $\sum_a (Qv)_a = 0$. Thus gamma acts on a direction's frequency content through the explicit weight A_γ^2 . Since $t/\sinh t$ decreases for $t > 0$, increasing gamma increases every directional response of H_γ and hence its ordered eigenvalues. This order concerns the integral kernel; exterior subtraction and normalization need not preserve monotonicity of every finite ratio.

A.2 TRANSFER TO THE FINITE DICTIONARY

For $u \perp \mathbf{1}$, the function $g(\zeta) = \sum_a u_a \tanh(\gamma(\zeta - z_a))$ is analytic for $|\text{Im } \zeta| < \pi/(2\gamma)$. Subtracting the common translate at $\bar{z}$ and integrating derivatives, Minkowski and Cauchy-Schwarz yield

$$\int_{\mathbb{R}} |g(t \pm i\theta/\gamma)|^2 dt \leq \frac{4\gamma}{3} \sec^4 \theta \|z - \bar{z}\mathbf{1}\|^2 \|u\|^2. \quad (16)$$

Here $|\operatorname{sech}^2(t + i\theta)| \leq \sec^2 \theta \operatorname{sech}^2 t$ and $\int \operatorname{sech}^4 t dt = 4/3$. Shift the contour for the analytic function $g(\zeta)^2$, not $|g(\zeta)|^2$: its transform is bounded by the right side of (16) times $e^{-\theta|\omega|/\gamma}$. Poisson summation at frequencies $2\pi k/h$ then gives

$$\left\| \frac{1}{M} \sum_{c \in c_0 + h\mathbb{Z}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q - H_\gamma \right\| \leq \delta_\gamma. \quad (17)$$

The geometric sum over $k \neq 0$ produces the denominator in (7).

Deleting exterior centers converts the infinite lattice to the actual dictionary. The retained deletions give T_γ . For a remaining center $c \geq c_R$, projection removes the constant limit and $1 - \tanh t \leq 2e^{-2t}$ gives an outer-product norm, divided by M , of at most $4e^{-4\gamma(c - z_{\max})}$. Summing this geometric series and its left-tail counterpart bounds the remaining positive deletion by $\tau_\gamma I$. This proves (9).

If κ_i are the eigenvalues of $Q^\top K_\gamma Q$, min-max and codimension-one interlacing give

$$\beta_i - \delta_\gamma - \tau_\gamma \leq \kappa_i \leq \beta_i + \delta_\gamma, \quad \mu_i \geq \kappa_i \geq \mu_{i+1}. \quad (18)$$

The Rayleigh quotient gives $\mu_1 \geq \ell_\gamma$. In the decomposition $\operatorname{span}(v_0) \oplus v_0^\perp$, the diagonal blocks of K_γ are bounded above by ℓ_γ and $c_\gamma I$, and the off-diagonal block has norm b_γ . Its quadratic form is therefore bounded by the largest eigenvalue of $\begin{bmatrix} \ell_\gamma & b_\gamma \\ b_\gamma & c_\gamma \end{bmatrix}$, namely L_γ . Dividing the interlacing bounds by these normalization endpoints proves the relative-rate enclosure from (12).

A.3 CLASSICAL OUTPUT-ERROR CONSEQUENCE

For $r_n = \Phi_\gamma w_n - y$, normalized least-squares GD gives $r_{n+1} = (I - \eta K_\gamma) r_n$ and $r_0 = -y$. Expansion in the actual finite-kernel eigenvectors gives

$$e_n^2 = \sum_i p_i (1 - \eta \mu_i)^{2n} = \sum_i p_i (1 - \rho_i/2)^{2n} \geq \sum_i p_i (1 - \bar{\rho}_i/2)^{2n}. \quad (19)$$

The last inequality follows because the factor decreases with $\rho_i \in [0, 1]$. This proves Theorem 3.2. Our step is a conservative fraction of the stability limit $2/\mu_1$; any nonoscillatory step $0 < \eta \mu_1 \leq 1$ instead uses $\eta \mu_1 \rho_i$. Only exact nullspace energy gives a permanent floor. For $\mu_i > 0$, $u_i = \Phi_\gamma(\Phi_\gamma^\top u_i / (M \mu_i))$ is representable, however small its rate. The target weights require the actual eigenvectors; the theorem bounds eigenvalues, not their target alignment.

B Evaluation, accuracy, and empirical scope

Which quantities are computed. The theorem endpoints use the explicit center integral, exterior-center subtraction, and scalar normalization bounds. A separate SVD of the actual finite feature matrix supplies the validation eigenvalues and target weights p_i . The plotted lower error uses the theorem's upper rates, not actual rates. In Figure 3 (left), ranks 4, 14, 30 carry approximately 46–48%, 8–14%, and 2.5–4.4% of sampled target energy, respectively, across the four evaluated bandwidths. The error bound uses all numerically resolved target projections.

The integral is evaluated through a positive quadrature factor rather than by subtracting nearly equal entries of (5). There is no Fourier truncation. Truncating the center integral at $\pm(\max_a |z_a| + P/\gamma)$ omits a positive matrix of norm at most $2e^{-4P}/(h\gamma)$, added to the numerical upper endpoint. The exterior remainder enters the lower endpoint. Compressing the combined integral and exterior factors Z to rank r introduces the two-sided bound $2\sigma_1(Z)\sigma_{r+1}(Z) + \sigma_{r+1}(Z)^2$ on their signed Gram matrix. Quadrature refinement and an empirical arithmetic guard are checked separately; the figures are FP64 evaluations, not interval-arithmetic certificates. Refining quadrature order from 10 to 16 and padding from 20 to 24 changes the lower error curves by at most 9.6×10^{-11} .

The remaining analytic slack comes from the interlacing index shift, the normalization bounds ℓ_γ , L_γ , and additive allowances. Small absolute errors can dominate extremely small eigenvalues; the enclosure is valid but need not be informative for every geometry or mode. In this experiment it is informative both for the plotted ratios and for the full error trajectory. Omitting unresolved target energy from the lower curve only weakens it; it is not interpreted as an exact approximation floor.

Target, geometry, and measurements. All displayed data use

$$f(x) = \frac{\sin(2\pi x) + \frac{1}{2}\sin(6\pi x) + \frac{1}{4}\sin(14\pi x)}{\sqrt{21/32}}, \quad x \in [-1, 1]. \quad (20)$$

The total hidden width is $W = 512$, comprising $N = 467$ interior cells and $R = 22$ halo centers on each side, so $h = 2/467$ and $W = N + 2R + 1$. There are 2,048 uniform training midpoints, 4,096 validation points, and 8,192 dense evaluation points. Every reported error is relative output L^2 error on its stated grid, not squared loss or an EMA. The dense grid checks resolution; it is not an untouched generalization test. Frozen GD starts its readout at zero and executes five million full-batch FP64 updates with $\eta = 1/(2\mu_1)$ at each bandwidth. Figure 3 (right) displays training error. Its spectral reference agrees with executed GD within 2.9×10^{-14} .

The joint-training comparison and feature interventions are documented in Appendix D. Neither the fixed-kernel proof nor its scalar GD rates apply to evolving features or Adam’s adaptive preconditioner.

Reproducibility. Figure sources, numerical provenance, protocols, and checks are indexed in `output/diagnostics/section34_long_horizon/README.md`. The figures reuse completed runs.

C Joint feature acquisition: conditions and full proof

The result controls an evolving tanh network in the ordinary Euclidean GD metric. A scalar comparison bounds the collective size of slopes, hidden biases, and readouts; a global remainder inequality converts that bound into output error. Its coefficients can be supplied as structural upper bounds or evaluated along a trajectory. The latter use is a conditional explanation, not a forecast from the initial parameters alone.

C.1 NETWORK, MEASURE, AND COARSE COMPENSATION

Let ν be a probability measure on $[-1, 1]$ with positive input variance, and write $\|\cdot\| = \|\cdot\|_{L^2(\nu)}$ throughout this appendix. For the training measure, this is the sample RMS norm $\|\cdot\|_{2,Z}$. Consider

$$\begin{aligned} q_\theta(x) &= d + \sum_{j=1}^W c_j \tanh(\gamma_j x + b_j), & L(\theta) &= \tfrac{1}{2} \|q_\theta - f\|^2, \\ \theta_{n+1} &= \theta_n - \eta_n \nabla L(\theta_n), & \eta_n &\geq 0. \end{aligned} \quad (21)$$

The target lies in $L^2(\nu)$ with $\|f\| > 0$, and $[a]_+ = \max(a, 0)$. Let P_C project onto affine functions, $P_H = I - P_C$, $f_\perp = P_H f$, $q_H = P_H q_\theta$, and $e_C = P_C(q_\theta - f)$, $e_H = q_H - f_\perp$. Jacobians $J_C = D_\theta(P_C q_\theta)$ and $J_H = D_\theta q_H$ map the Euclidean parameter space into these output spaces. Their adjoints use the stated output measure; empirical adjoints therefore include the sample normalization.

At the iterates under consideration assume $K_C = J_C J_C^*$ is invertible on the two-dimensional affine space. Define

$$\begin{aligned} \Pi &= I - J_C^* K_C^{-1} J_C, & \ell &= K_C^{-1} J_C J_H^* e_H, \\ F &= \Pi J_H^* e_H, & R &= J_C^* (e_C + \ell). \end{aligned} \quad (22)$$

Then $\nabla L = F + R$, Π is an orthogonal projection, and $J_C F = 0$. Thus F includes the compensation needed to preserve affine output to first order. Small tracking R does not mean that compensation vanishes, or that e_C is exactly zero. The proof uses $\|\Pi\| \leq 1$; it needs no quantitative lower singular-value bound on J_C . At a singular coarse Gram matrix this formulation is not invoked.

Write $\vartheta = (\gamma, b, c)$ for the hidden parameter blocks, omitting the output bias, and set

$$r_j = (\gamma_j^2 + b_j^2 + c_j^2)^{1/2}, \quad S_k = \sum_j r_j^k, \quad s = \|\vartheta\|_2 = \sqrt{S_2} > 0. \quad (23)$$

The slope and center parameters are free to evolve. Here b_j is the hidden intercept; a center, when $\gamma_j \neq 0$, is $-b_j/\gamma_j$. No bound on center displacement is inferred by dividing by a small slope.

C.2 GLOBAL TANH AND DERIVATIVE REMAINDER BOUNDS

Put $u_j = \gamma_j x + b_j$ and define

$$\Psi_3 = -\frac{1}{3} \sum_j c_j u_j^3, \quad \Psi_5 = \frac{2}{15} \sum_j c_j u_j^5, \quad J_k = D_\theta(P_H \Psi_k), \quad G_k = J_k^* f_\perp \quad (k = 3, 5). \quad (24)$$

Set $t_3 = 1/3$, $t_5 = 2/15$, $t_7 = 17/315$ and

$$A_k = t_k \frac{2^{k/2} k^{k/2}}{(k+1)^{(k+1)/2}}, \quad C_k = (k+1)A_k, \quad k = 3, 5, 7. \quad (25)$$

In particular $A_3 = \sqrt{6}/8$. For every real u , the remainders after the linear, cubic, and quintic tanh polynomials satisfy

$$|\mathcal{R}_k(u)| \leq t_k |u|^k, \quad |\mathcal{R}'_k(u)| \leq k t_k |u|^{k-1}, \quad k = 3, 5, 7. \quad (26)$$

Here $\mathcal{R}_3 = \tanh u - u$, $\mathcal{R}_5 = \tanh u - u + u^3/3$, and $\mathcal{R}_7 = \tanh u - u + u^3/3 - 2u^5/15$.

Proof of the scalar bounds. Since $|\tanh u| \leq |u|$, $|\mathcal{R}'_3| = \tanh^2 u \leq u^2$; integration gives the cubic bound. Next, $\mathcal{R}'_5 = u^2 - \tanh^2 u$, whose magnitude is at most $|u - \tanh u| |u + \tanh u| \leq 2|u|^4/3$; integrate again. For the last pair, $\sup_{u \in \mathbb{R}} |\tanh^{(7)} u| \leq 272$. To check this constant explicitly, write $z = \tanh^2 u \in [0, 1]$:

$$\tanh^{(7)} u = -272 + 3968z - 12096z^2 + 13440z^3 - 5040z^4.$$

The degree-four Bernstein coefficients on $[0, 1/2]$ and $[1/2, 1]$ are $(-272, 224, 216, 124, 53)$ and $(53, -18, -68, 8, 0)$, respectively. The Bernstein basis is nonnegative and sums to one, proving the bound. Taylor's integral remainder for $\tanh$ through degree six and its derivative through degree five gives $272/7! = 17/315$ and $272/6! = 17/45$. The intervening coefficients vanish by parity.

From neurons to output and Jacobians. On $[-1, 1]$, $|u_j| \leq \sqrt{2}(\gamma_j^2 + b_j^2)^{1/2}$. At fixed r_j , maximizing $t_k 2^{k/2} |c_j| (\gamma_j^2 + b_j^2)^{k/2}$ gives $A_k r_j^{k+1}$. The squared norm of that neuron's three derivative columns is bounded pointwise by

$$t_k^2 2^k [k^2 c_j^2 (\gamma_j^2 + b_j^2)^{k-1} + (\gamma_j^2 + b_j^2)^k] \leq C_k^2 r_j^{2k}.$$

The maximum occurs at $(\gamma_j^2 + b_j^2)/r_j^2 = k/(k+1)$. Sum output bounds and squared column bounds, integrate against the probability measure, and use contractivity of P_H . With the Hilbert–Schmidt norm bounding the operator norm, this proves

$$\begin{aligned} \|q_H - P_H(\Psi_3 + \Psi_5)\| &\leq A_7 S_8, \\ \|J_H - J_3 - J_5\|_{\text{HS}} &\leq C_7 \sqrt{S_{14}}, \\ \|J_H^* f_\perp - G_3 - G_5\|_2 &\leq C_7 \|f_\perp\| \sqrt{S_{14}}. \end{aligned} \quad (27)$$

The same argument gives $\|q_H\| \leq A_3 S_4$ and $\|J_H\|_{\text{HS}} \leq C_3 \sqrt{S_6}$. These are global inequalities for the exact tanh network: no neuronwise small-argument hypothesis or polynomial replacement of training is used.

C.3 STRUCTURAL COEFFICIENTS AND TARGET DEPENDENCE

For $s > 0$ define

$$\begin{aligned}\chi_k &= W^{k/2-1} S_k / s^k, \\ \alpha_k &= W^{(k-1)/2} \|P_H \Psi_k\| / s^{k+1}, \\ \beta_k &= W^{(k-1)/2} \|J_k\|_{\text{HS}} / s^k, \quad \zeta_k = W^{(k-1)/2} \|G_k\|_2 / s^k, \quad k = 3, 5.\end{aligned}\tag{28}$$

These coefficients describe concentration, polynomial output, sensitivity, and target coupling. They are invariant under common positive rescaling of ϑ ; bounding them does not itself bound s . For equal neuron norms, $\chi_k = 1$; for the moments $k > 2$ used here, concentration in fewer neurons increases it. Let $\tau_k = \|\mathcal{P}_k f_\perp\|$, where $\mathcal{P}_k$ is the space of polynomials of degree at most k for the same measure. Then

$$\zeta_k \leq \beta_k \tau_k, \quad \alpha_k \leq A_k \chi_{k+1}, \quad \beta_k \leq C_k \sqrt{\chi_{2k}}, \quad k = 3, 5.\tag{29}$$

Each column of J_k is a polynomial of degree at most k , so $J_k^* f_\perp = J_k^* \mathcal{P}_k f_\perp$. The other inequalities follow from the fixed-radius bounds just proved. Thus cubic target orthogonality makes $G_3 = 0$, and quintic orthogonality also makes $G_5 = 0$. The mixed-sine and localized targets generally have nonzero low-degree coupling. The numerical evaluations retain $\alpha_k, \beta_k, \zeta_k$; substituting their concentration-only bounds gives a different, potentially looser evaluation, whose tightness is not claimed here.

At each GD iterate n , take finite nonnegative upper bounds $\hat{\alpha}_{k,n}, \hat{\beta}_{k,n}, \hat{\zeta}_{k,n}, \hat{\chi}_{8,n}, \hat{\chi}_{14,n}$ on these coefficients, and $\hat{r}_n \geq \|R_n\|_2$. Exact coefficients are allowed. For a candidate radius $B \geq 0$, define

$$\begin{aligned}\mathcal{Q}_n(B) &= \hat{\alpha}_{3,n} \frac{B^4}{W} + \hat{\alpha}_{5,n} \frac{B^6}{W^2} + A_7 \hat{\chi}_{8,n} \frac{B^8}{W^3}, \\ \mathcal{J}_n(B) &= \hat{\beta}_{3,n} \frac{B^3}{W} + \hat{\beta}_{5,n} \frac{B^5}{W^2} + C_7 \sqrt{\hat{\chi}_{14,n}} \frac{B^7}{W^3}, \\ \mathcal{T}_n(B) &= \hat{\zeta}_{3,n} \frac{B^3}{W} + \hat{\zeta}_{5,n} \frac{B^5}{W^2} + C_7 \|f_\perp\| \sqrt{\hat{\chi}_{14,n}} \frac{B^7}{W^3}, \\ \mathcal{V}_n(B) &= \mathcal{J}_n(B) \mathcal{Q}_n(B) + \mathcal{T}_n(B).\end{aligned}\tag{30}$$

For $s_n \leq B$, these bound $\|q_{H,n}\|, \|J_{H,n}\|, \|J_{H,n}^* f_\perp\|_2$, and $\|F_n\|_2$, respectively. For the last,

$$\|F_n\|_2 \leq \|J_{H,n}\| \|q_{H,n}\| + \|J_{H,n}^* f_\perp\|_2 \leq \mathcal{V}_n(B).\tag{31}$$

All four polynomials are nondecreasing in B . Their coefficients may evolve; neither the Jacobian nor the target coupling is frozen.

C.4 PRECISE GD THEOREM AND PROOF

Theorem C.1 (Population growth and output error for joint GD). *Consider (21) from an arbitrary restart, indexed by $n = 0$. Suppose $K_{C,n}$ is invertible, $s_n > 0$, and the structural upper bounds above hold for every iterate $0 \leq n \leq N$. Define*

$$B_0 = s_0, \quad B_{n+1} = B_n + \eta_n [\mathcal{V}_n(B_n) + \hat{r}_n].\tag{32}$$

Whenever these quantities are finite, for $0 \leq n \leq N$,

$$s_n \leq B_n, \quad \sum_{k < n} \|\vartheta_{k+1} - \vartheta_k\|_2 \leq B_n - B_0,\tag{33}$$

$$\frac{\|f - q_n\|}{\|f\|} \geq \frac{[\|f_\perp\| - \mathcal{Q}_n(B_n)]_+}{\|f\|}, \quad \bar{\lambda}_n \leq \lambda_{\text{RMS},n} \leq \frac{h B_n}{\sqrt{W}}.\tag{34}$$

Here $h > 0$ is the reference spacing, $\lambda_{\text{RMS},n} = h\|\gamma_n\|_2/\sqrt{W}$, and $\bar{\lambda}_n = hW^{-1}\sum_j |\gamma_{j,n}|$.

Proof. Initially $s_0 = B_0$. If $s_n \leq B_n$, the exact GD update and (31) give

$$\|\vartheta_{n+1} - \vartheta_n\|_2 \leq \eta_n \|F_n + R_n\|_2 \leq \eta_n [\mathcal{V}_n(B_n) + \hat{r}_n] = B_{n+1} - B_n.$$

The triangle inequality implies $s_{n+1} \leq B_{n+1}$; induction and summation prove (33). Orthogonal projection and the reverse triangle inequality yield

$$\|q_n - f\| \geq \|q_{H,n} - f_\perp\| \geq [\|f_\perp\| - \|q_{H,n}\|]_+ \geq [\|f_\perp\| - \mathcal{Q}_n(B_n)]_+.$$

Finally Cauchy–Schwarz and $\|\gamma_n\|_2 \leq s_n$ prove both bandwidth bounds. These arguments use the discrete updates directly, with no continuous-time approximation or unmeasured finite-step correction. $\square$

The result holds for any nonnegative step sizes for which its premises hold. Large steps can make the comparison uninformative. Small post-transient tracking makes it useful but is not silently set to zero. A large value of B_n or a zero output floor signals loss of this sufficient explanation; it does not establish successful acquisition. Coefficient bounds checked only at stored times do not certify their values at every intervening GD step. For a pooled mean across equal-sized seeds, the per-seed bounds average; the theorem does not interpret neuron spread as uncertainty across seeds.

D Evidence and interpretation for Section 3.5

D.1 FIVE-MILLION-UPDATE OBSERVATION AND FEATURE ASSAYS

The primary target is $[\sin(2\pi x) + \frac{1}{2}\sin(6\pi x) + \frac{1}{4}\sin(14\pi x)]/\sqrt{21/32}$ on $[-1, 1]$. Total width is 512, including 22 halo nodes per side; the reference spacing is $h = 2/467$. Full-batch FP64 training uses 2,048 midpoints and five paired initializations. One recipe per joint optimizer is selected by median final error on 4,096 validation midpoints, requiring finite endpoints for every seed. The 8,192-point evaluation grid is a previously inspected resolution check, not an untouched test set. The selected joint Adam and GD recipes use initial rates 0.05 and 0.2, respectively, with cosine decay over all five million updates. The supplied-feature Adam comparison uses $\lambda = 1/4$, zero readout initialization, and the selected full-horizon cosine rate 10^{-4} . Both constant and cosine candidates are retained in the study records.

Median joint training errors are 1.8143227×10^{-3} for Adam and 0.24393137 for GD. Supplied-feature Adam reaches 6.6245807×10^{-7} . The corresponding dense-grid values are 1.8148250×10^{-3} , 0.24393487 , and 6.6198225×10^{-7} . Joint Adam’s median endpoint error falls about 28% between the two- and five-million-update experiments. This is a precision gap within the stated budget, not permanent failure. The pooled endpoint mean bandwidths are 0.0168845 for Adam and 0.00135795 for GD. Their neuron population standard deviations are 0.0541204 and 0.00703476. The supplied bandwidth is a comparison geometry, not a target-independent necessary threshold. The revised three-panel Figure 4 is external to this review copy; no new width-scaling result is asserted here.

Freezing the five-million-update Adam features and training new readouts for two million additional updates gives median dense-grid error 1.8147369×10^{-3} . SVD readout fits at relative cutoffs 10^{-12} and 10^{-14} also yield only small improvements; lowering the cutoff to 10^{-16} exposes sensitivity to floating-point cancellation. These assays support a limitation of the learned dictionary at the checked numerical precision, not an exact-arithmetic approximation floor. Multiplying both slope and intercept by 4 or 16 preserves each learned center. After the same readout-training assay, median dense-grid errors are 0.0916060 and 0.138935, respectively. Thus simple slope amplification does not repair this geometry. These interventions change nonnested feature spaces; they do not establish that larger slopes always help. The assay’s additional optimization budget is separate from the end-to-end five-million-update comparison.

D.2 CONDITIONAL THEOREM EVALUATION ON EVOLVING GD POPULATIONS

The mechanism study uses a separate cohort: total width 705, reference spacing $h = 1/256$, 2,048 uniform midpoint samples, and restarts after 20,000 ordinary-GD updates at step size 0.002. Each continuation runs another 100,000 updates, or 200 units of flow time. Seed 30 covers six targets: an orthogonal degree-five mixture, mixed sine, a shifted Gaussian, a compact bump, a smooth step, and an absolute-value kink. The mixture is $0.3P_0 + 0.4P_1 + \sqrt{0.75}P_5$, with P_k orthonormal for the empirical measure. The other unnormalized targets are

$$\begin{aligned} & \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(14\pi x), \\ & e^{-((x+0.35)/0.22)^2}, \quad \tanh(14(x-0.31)), \quad |x+0.23|, \\ & f_{\text{bump}}(x) = \begin{cases} \exp\left(1 - \frac{1}{1-u^2}\right), & |u| < 1, \quad u = (x-0.35)/0.22, \\ 0, & |u| \geq 1. \end{cases} \end{aligned}$$

Each is divided by its RMS on the original training grid. Four additional seed-31 restarts cover step, kink, Gaussian, and bump; they were selected for weak initial comparison margins before their continuation outcomes were inspected.

The discrete scalar recurrence uses structural diagnostics sampled every unit of flow time, interpolating its nonnegative polynomial coefficients and tracking norm to native GD steps. This implements a numerical evaluation of the conditional theorem; interpolation is not a proved upper enclosure between stored samples. In the six principal GD runs the minimum sampled relative-error floors are, in the target order above, 0.865734, 0.912194, 0.784451, 0.879212, 0.479360, and 0.426952. The corresponding executed endpoint errors are 0.866025, 0.912564, 0.788073, 0.881027, 0.489675, and 0.427513. The minimum floors and endpoint errors are distinct summaries, not an every-step tightness statistic. All ten growing comparisons remain useful over the full continuation, with seed-31 minimum floors from 0.426695 to 0.878467. Principal endpoint RMS-bandwidth bounds range from 3.63×10^{-4} to 5.18×10^{-4} , versus observed values from 2.09×10^{-4} to 2.51×10^{-4} .

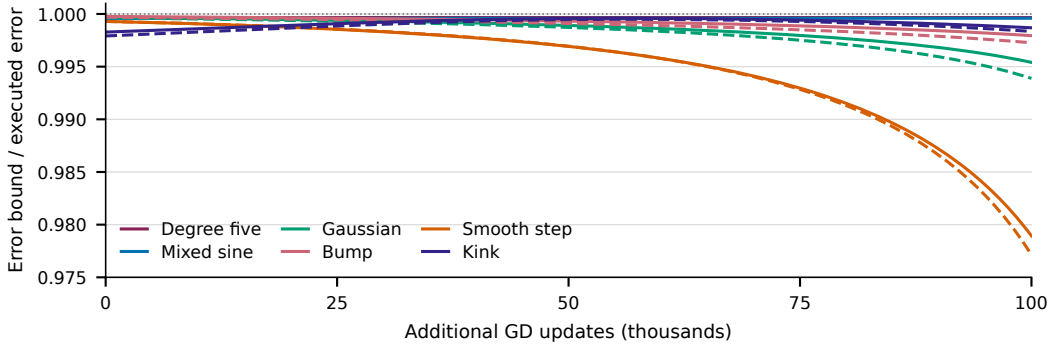


Figure 6: **The output-error bound remains informative across all ten continuations.** The theorem’s lower bound divided by executed relative training RMS error at each stored diagnostic time, every 500 GD updates. Solid curves are seed 30; dashed curves are the four seed-31 stress cases. The smallest ratio is 0.97714. These are evaluations with interpolated structural coefficients from the trajectories, not initial-state forecasts or certified bounds between diagnostic samples.

Figure 6 compares bound and error at matching times throughout the trajectories. The relative shortfall never exceeds 2.29% at stored diagnostic times. This statistic directly measures output-bound tightness, separately from the endpoint summaries above.

The saved verification includes independent automatic derivatives, global remainder checks, exact discrete moment identities, and a solvable positive-growth comparison. Step refinement changes population size and normalized slope RMS by less than 4.8×10^{-7} relatively across both seeds.

Decimating the structural diagnostics changes the final GD comparison radius by less than 9×10^{-6} relatively. These checks support numerical resolution; they do not supply interval-arithmetic certificates for the conditional premises.

The theorem explains this post-transient regime through evolving population structure. The older width-177 archive includes regimes where large moments make the polynomial comparison uninformative. No coverage claim for all 23 archived targets, all training ages, or the full five-million-update Figure 4 trajectories follows. Adam’s coordinate-dependent scaling and momentum also fall outside the ordinary-GD proof. Its observed precision gap motivates the same empirical question without establishing the same quantitative mechanism.

References

- Matthew Tancik, Pratul Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, Utkarsh Singhal, Ravi Ramamoorthi, Jonathan Barron, and Ren Ng. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL https://papers.neurips.cc/paper_files/paper/2020/hash/55053683268957697aa39fba6f231c68-Abstract.html.
- Zhi-Qin John Xu, Yaoyu Zhang, Tao Luo, Yanyang Xiao, and Zheng Ma. Frequency principle: Fourier analysis sheds light on deep neural networks. *Communications in Computational Physics*, 28(5):1746–1767, 2020. doi: 10.4208/cicp.OA-2020-0085. URL <https://www.global-sci.com/cicp/article/view/6896>.